

Alteryx Assignment 3

Using Alteryx to Audit Accounting Data¹

Instructions

This exercise is about accounts receivable aging analysis. Before working on this assignment, you should review the concepts of accounting treatment of cash and accounts receivable.

Objective

You can use Alteryx to aggregate and analyze accounting data, which you can then write to an Excel spreadsheet. The following examples demonstrate a common accounting task, creating an aging analysis of accounts receivable.

Deliverable

1. Write a report based on what you did in **each part** of this assignment.
2. **You need to comment on the findings** as you were the auditor.
3. Include the screenshot(s) of the **critical** workflow in Alteryx and/or the output in Excel that you will use to assist the readers in following your logic while reading your report.
4. Convert the Word file into a PDF file for submission.
5. Name it **A455-Alteryx3-byYourName** and upload the PDF file on Canvas.
6. In taking the screenshots of both your workflow and your Excel outputs, make sure that these are full screens like you did in the two previous Alteryx exercises.

IMPORTANT: Remember to include a **comment box** with your name on **EACH Alteryx** screenshot (have the comment box visible in the background of each screenshot).

¹ Please allocate 2 to 2.5 hours to complete this assignment.

Part 1: Create a one-dimension aging analysis

Please download the data file called “AR data” to be used for this Alteryx exercise. The “breakdowns of sales” spreadsheet (the first tab in the data file) contains the purchase records for 107 fictitious unpaid invoices. Each record includes information such as customer number, customer type, assigned salesperson, account balance (i.e., Bal.), and invoice date (i.e., Inv. Date). Additionally, the information on customer type is given in the “customer type” second tab sheet.

What a manager or auditor might desire is an aging report that summarizes this data to assess the collectability of customer accounts.

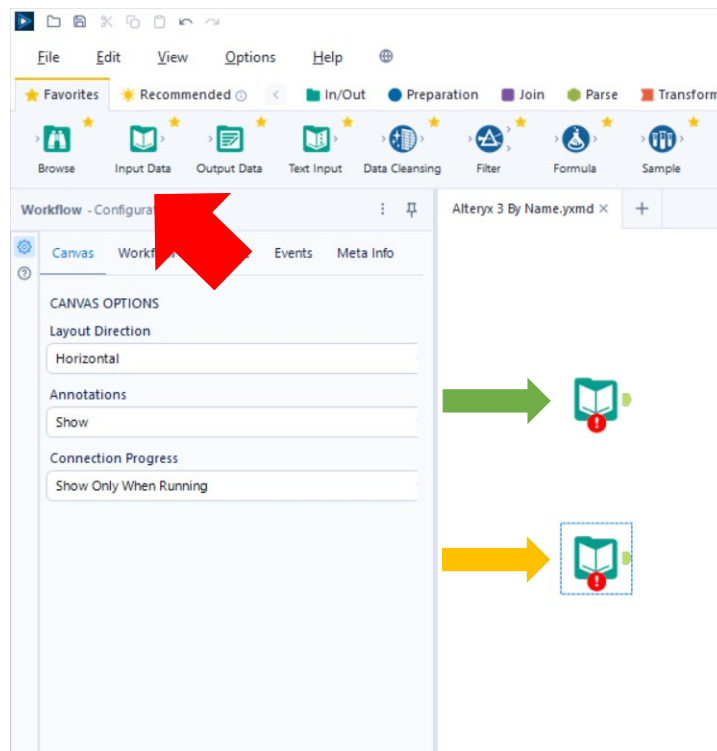
For this assignment, consider the date as “2/1/2025”.

	A	B	C	D	E	F
	Customer	sales person	Bal.	Inv. Date		
2	3	C	\$ 145.33	1/20/2025		
3	5	C	\$ 49.32	11/16/2024		
4	3	D	\$ 110.88	12/16/2024		
5	2	D	\$ 50.81	1/25/2025		
6	8	B	\$ 61.69	1/15/2025		
7	5	C	\$ 108.38	12/31/2024		
8	4	D	\$ 142.72	12/9/2024		
9	4	E	\$ 60.04	1/11/2025		
10	7	D	\$ 90.04	1/20/2025		
11	2	A	\$ 20.61	1/17/2025		
12	2	A	\$ 147.19	1/18/2025		
13	9	D	\$ 10.59	11/18/2024		
14	5	C	\$ 48.92	1/18/2025		
15	8	E	\$ 25.11	12/30/2024		
16	9	D	\$ 31.72	11/27/2024		

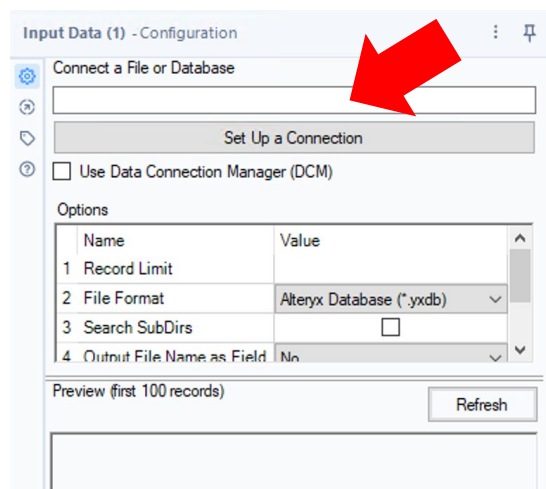
Sample Accounts Receivable Data (AR Data.xlsx)

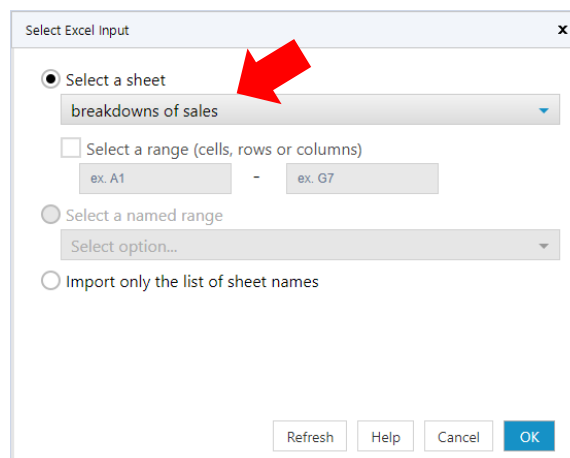
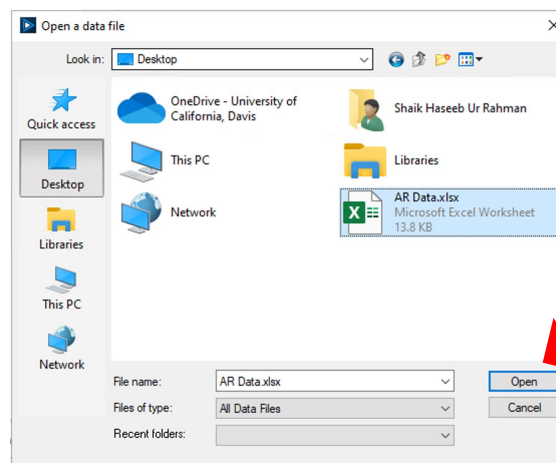
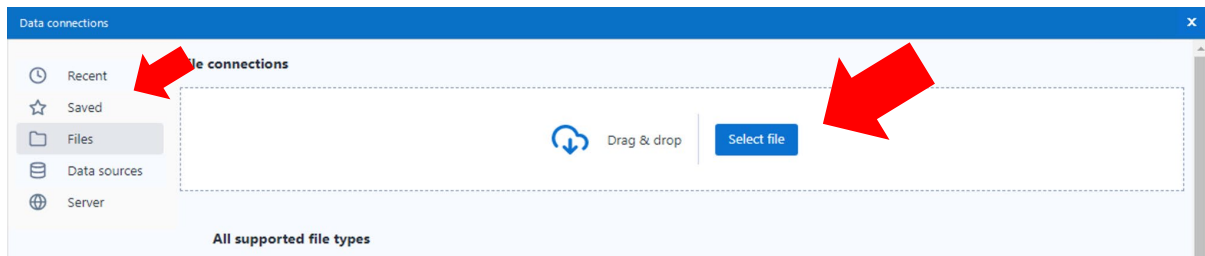
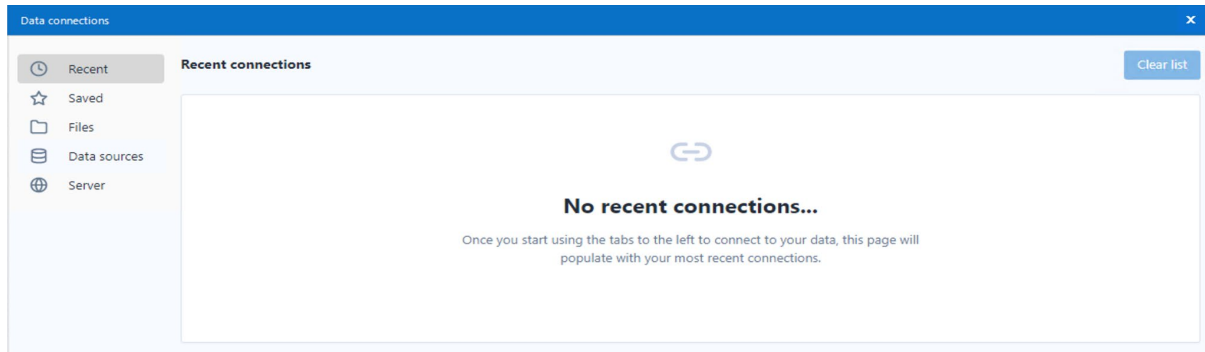
1. Create aging data and classification

- A) Click on your 'Favorites' or the 'In/Out' in the toolbar, and drag the tool called 'Input Data' onto the canvas two times. There will be a "top" or "upper" flow and a "bottom" or "lower" flow when distinguishing between the two flows.



- B) For the input data on the **upper flow** (refer to figure 1A, green arrow), configure the input data tool to pull data from the "AR Data.xlsx" file on your computer and select the sheet "**breakdowns of sales**." Click on Set Up a connection -> Files -> Select File.





- C) Do the same thing for the input data on the **lower flow** (refer to figure 1A, yellow arrow), but this time click on the second tab sheet **“Customer Type.”**

Select Excel Input

Select a sheet

customer type

Select a range (cells, rows or columns)

ex. A1

-

ex. G7

Select a named range

Select option...

Import only the list of sheet names

Refresh

Help

Cancel

OK

- D) Now we want to join this data together in the Excel sheet. Drag one **'Join'** tool from 'Favorites' or 'Join' in the toolbar onto the canvas and connect it to both Input Data tools. Think about what these two sheets have in common. In the configuration tab, click on 'Customer' and then run the workflow. Change the **Type** for both the **'Customer'** fields to **V_String**. (**Hint:** You will need to change the format of the customer number from "general" to "text" **before** importing the file to Alteryx for processing. Otherwise, you will get a warning message in Alteryx.)

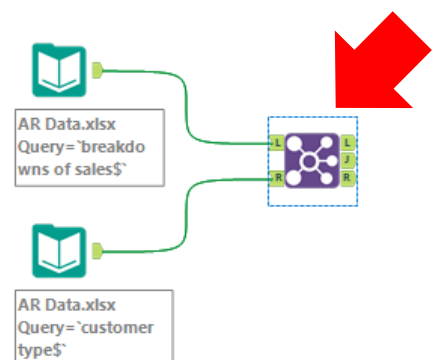
Join (3) - Configuration

☒ Join by Record Position
☐ Join by Specific Fields

	Left		Right
1	Customer	↔	Customer
*		↔	

Options: ☐ Left ☐ Right

	Input	Field	Type	Rename	Desc
☒	Left	Customer	V_String		
☒	Left	sales person	V_String	255	
☒	Left	Bal.	Double	8	
☒	Left	Inv. Date	Date		
☒	Right	Customer	V_String		Right_Customer
☒	Right	Cust. Type	V_String	255	
☒		*Unknown	Unknown	0	



E) You can now see that the columns are conjoined, but there is an extra column that we don't want.

Results - Formula (24) - Output

7 of 7 Fields | 1,649 records displayed

Record	Customer	sales person	Bal.	Inv. Date	Right_Customer	Cust. Type	Aging
1	2	D	50.81	2025-01-25	2	X	7
2	2	D	50.81	2025-01-25	2	X	7
3	2	D	50.81	2025-01-25	2	X	7
4	2	D	50.81	2025-01-25	2	X	7
5	2	D	50.81	2025-01-25	2	X	7
6	2	D	50.81	2025-01-25	2	X	7
7	2	D	50.81	2025-01-25	2	X	7
8	2	D	50.81	2025-01-25	2	X	7

To get rid of the “Right_Customer” column, unclick that category in the configuration tab. Then click on “Join by record position” and run the workflow.

Alteryx Designer x64 - *Alteryx 3 By Name.yxmd

File Edit View Options Help

Favorites Recommended In/Out Preparation Join Parse Transform In-Database Reporting Documentation Spatial Machine Learning

Browse Input Data Output Data Text Input Data Cleansing Filter Formula Sample Select Sort Join Union Text To Columns Summarize Comment

Join (3) - Configuration

Join by Record Position

Join by Specific Fields

Left Right

Options

Input	Field	Type	Size
Left	Customer	V_String	214...
Left	sales person	V_String	255
Left	Bal.	Double	8
Left	Inv. Date	Date	10
Right	Customer	V_String	214...
Right	Cust. Type	V_String	255
*Unknown	Unknown	Unknown	0

AR Data.xlsx
Query='breakdowns of sales\$'

AR Data.xlsx
Query='customer type\$'

F) Now we want to add an “Aging” column because we want to know how many days have passed since each of the invoice dates. We can do this by using the formula tool. Connect the ‘Formula’ tool from ‘Favorites’ to the join tool and add a new column called “Aging”.

G) Use the function ‘DateTimeDiff’ to subtract the invoice dates from the current date (in this case we are using 2/1/2025 as the current date). Change the data type to Double.

- It should look like: **DateTimeDiff('2025-02-01',[Inv. Date],'days')**

Formula (24) - Configuration

Output Columns: Aging

Formula: `DateTimeDiff('2025-02-01', [Inv. Date], 'days')`

Data type: Double, Size: 8

Workflow Diagram:

- AR Data.xlsx (Query="breakdowns of sales\$")
- AR Data.xlsx (Query="customer type\$")
- Join tool
- Formula tool (Aging = DateTimeDiff('2025-02-01', [Inv. Date], 'days'))

ACC 455
Alteryx #3
Solution!

H) Then run the workflow. You can see that the new column has been added.

Results - Formula (24) - Output

6 of 6 Fields | Cell Viewer | 107 records displayed | Search

Record	Customer	sales person	Bal.	Inv. Date	Cust. Type	Aging
1	3	C	145.33	2025-01-20	Y	12
2	5	C	49.32	2024-11-16	Y	77
3	3	D	110.88	2024-12-16	Y	47
4	2	D	50.81	2025-01-25	X	7
5	8	B	61.69	2025-01-15	Z	17
6	5	C	108.38	2024-12-31	Y	32
7	4	D	142.72	2024-12-09	Y	54
8	4	E	60.04	2025-01-11	Y	21

I) To classify the customer records into standard categories such as **“current,” “30-59 days,” “60-89 days,”** and **“over 90 days,”** we need to add a new column. We will classify it like this:

0	current
30	30-59 days
60	60-89 days
90	over 90 days

- J) Drag the **Formula** tool to the canvas and connect it to the previous Formula tool. Add a column named '**Classification**' and think about what formula to use. A conditional statement would get us the result that we need, so type in the formula:

```
IF [Aging]<30 THEN 'current'
ELSEIF [Aging]>=30 and [Aging]<60 THEN '30-59 days'
ELSEIF [Aging]>=60 and [Aging]<90 THEN '60-89 days'
ELSE 'over 90 days'
ENDIF
```

ACC 455
Alteryx #3
Solution!

- K) Run the workflow. You can see that a new column named Classification has been added.

Record	Customer	sales person	Bal.	Inv. Date	Cust. Type	Aging	Classification
1	3	C	145.33	2025-01-20	Y	12	current
2	5	C	49.32	2024-11-16	Y	77	60-89 days
3	3	D	110.88	2024-12-16	Y	47	30-59 days
4	2	D	50.81	2025-01-25	X	7	current
5	8	B	61.69	2025-01-15	Z	17	current
6	5	C	108.38	2024-12-31	Y	32	30-59 days
7	4	D	142.72	2024-12-09	Y	54	30-59 days
8	4	E	60.04	2025-01-11	Y	21	current

Now drag the  **Select** tool from the '**Favorites**' toolbar onto the canvas and connect it to the formula tool. Change the "**Inv Date**" type to a String.

Select (6) - Configuration

Options | ↑ ↓ | TIP: To reorder multiple rows: select, right-click and drag.

	Field	Type	Size	Rename	Description
☒	Customer	V_String	19		
☒	sales person	V_String	255		
☒	Bal.	Double	8		
☒	Inv. Date	String	10		
☒	Cust. Type	V_String	255		
☒	Aging	Double	8		
☒	Classification	V_WString	107...		
☒	*Unknown	Unknown	0		Dynamic or Unkno...

Select (6) - Configuration

Options | ↑ ↓ | Search

	Field	Type	Size	Rename
☒	Customer	V_String	214...	
☒	sales person	V_String	255	
☒	Bal.	Double	8	
☒	Inv. Date	String	16384	
☒	Cust. Type	V_String	255	
☒	Aging	Double	8	
☒	Classification	V_WString	107...	
☒	*Unknown	Unknown	0	

*Alteryx 3 By Name.xymd

AR Data.xlsx
Query='breakdo
wins of sales'

AR Data.xlsx
Query='customer
type\$'

Aging =
DateTimeDiff
('2023-02-01',
[Inv. Date], days)

Classification = IF
[Aging] < 30 THEN
'current'
ELSEIF [Aging]
>= 30 and [Aging]
< 60...

L) Drag the 'Block until done' tool from 'Developer' to the canvas and connect it to the Select tool.

Alteryx Designer x64 - *Alteryx 3 By Name.xymd

File Edit View Options Help

Search for tools, help, and resources...

Developer Laboratory Connect SDK Examples

Block Until Done (7) - Configuration

This tool has no configuration

*Alteryx 3 By Name.xymd

AR Data.xlsx
Query='breakdo
wins of sales'

AR Data.xlsx
Query='customer
type\$'

Aging =
DateTimeDiff
('2023-02-01',
[Inv. Date], days)

Classification = IF
[Aging] < 30 THEN
'current'
ELSEIF [Aging]
>= 30 and [Aging]
< 60...

- M) Now we need to output the data into an Excel file. Drag the **'Output Data'** tool and connect it to the **#1** outlet anchor of the Block Until Done tool. In the configuration panel, find the 'Set Up a Connection' dropdown and select the file to save to. Specify it as **Sheet1** and click "Overwrite file (remove)" in the configuration tab.

Output Data (8) - Configuration

Write to File or Database

C:\Users\hshaik00\Desktop\Alteryx#3Soln.xlsx||Sheet1

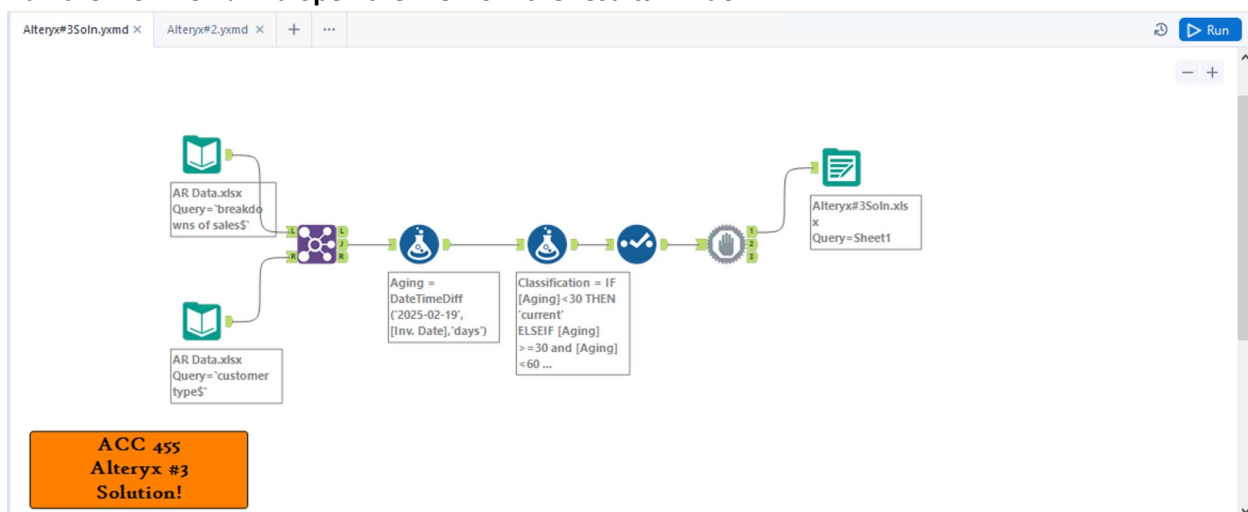
Set Up a Connection

☐ Use Data Connection Manager (DCM)

Options

Name	Value
1 Max Records Per File	
2 File Format	Microsoft Excel (*.xlsx) ▾
3 Output Options	Overwrite File (Remove) ▾
4 Append Field Map	By Field Name ...
5 Skip Field Names	☐
6 Preserve Formatting on Overwrite (Range Required)	☐
7 Enable SQL Server FileTable Support	☐

Run the workflow! And open the file from the results window.



O) The output file will be something like:

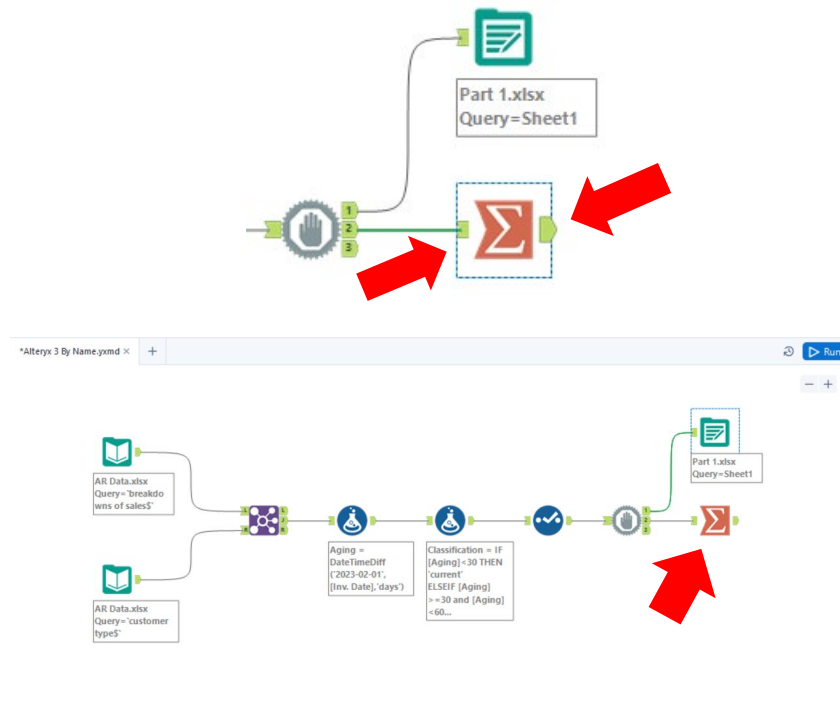
	A	B	C	D	E	F	G
1	Customer	sales pers	Bal.	Inv. Date	Cust. Type	Aging	Classification
2	3	C	145.33	2025-01-20	Y	12	current
3	5	C	49.32	2024-11-16	Y	77	60-89 days
4	3	D	110.88	2024-12-16	Y	47	30-59 days
5	2	D	50.81	2025-01-25	X	7	current
6	8	B	61.69	2025-01-15	Z	17	current
7	5	C	108.38	2024-12-31	Y	32	30-59 days
8	4	D	142.72	2024-12-09	Y	54	30-59 days
9	4	E	60.04	2025-01-11	Y	21	current
10	7	D	90.04	2025-01-20	Y	12	current
11	2	A	20.61	2025-01-17	X	15	current
12	2	A	147.19	2025-01-18	X	14	current
13	9	D	10.59	2024-11-18	Z	75	60-89 days
14	5	C	48.92	2025-01-18	Y	14	current
15	8	E	25.11	2024-12-30	Z	33	30-59 days
16	9	D	31.72	2024-11-27	Z	66	60-89 days
17	4	A	16.83	2025-01-05	Y	27	current

Figure 1: Sample Accounts Receivable Data (Output)

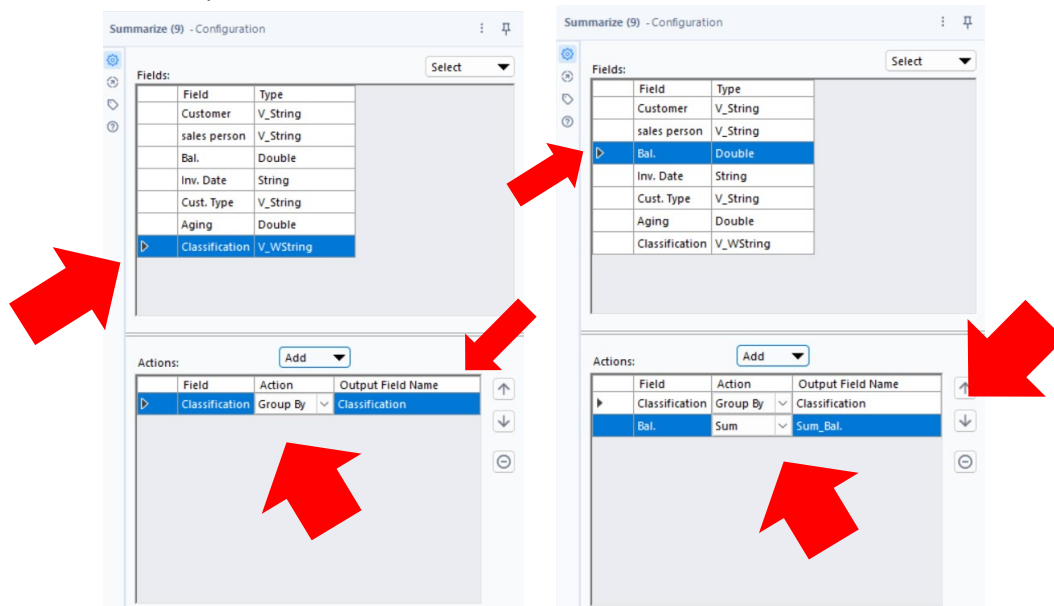
2. Create a table to perform Aging analysis

If we want to have a pivot table in our Excel sheet, we can simply use the Summarize tool in Alteryx to do that. This will help us compute the desired aging analysis.

- A) Select the **Summarize** tool from the 'Favorites' tool and connect it to #2 of the **Block Until Done** tool.



- B) Then in the configuration tab, add '**Classification**' and press **Group By**. Then add '**Bal.**' and press **Sum**.



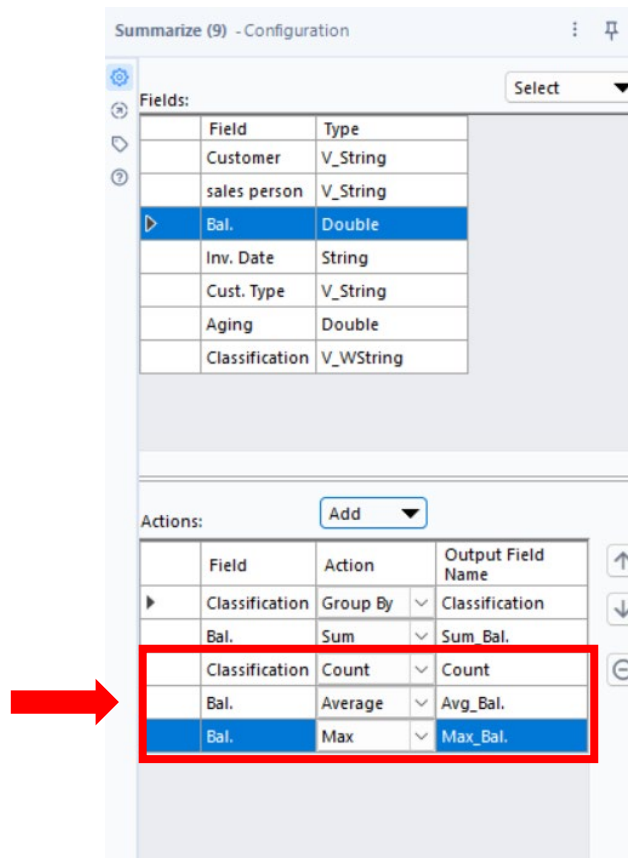
If you run the workflow and click 'Output' in the results panel, you'll see a small table showing the total balances of the different classification groups.

Results - Summarize (53) - Output

2 of 2 Fields ☒ ☐ Full Viewer

Record	Classification	Sum_Bal.
1	30-59 days	2457.36
2	over 90 days	3338.38
3	current	3189.24
4	60-89 days	1637.12

- C) The auditor may also want values other than (or in addition to) the sum of the amounts owing for each category in an aging analysis. For example, suppose that the auditor wants to examine
- 1) the sum of the transaction amounts for each aging category,
 - 2) a count of the number of transactions,
 - 3) the average of the transaction balances for each aging category, and
 - 4) the maximum transaction amount for each category. This is easily accomplished by using the Summarize tool.
- D) To perform this task, add '**Classification**' and press '**Count.**' Then add '**Bal.**' and click on '**Numeric**' then '**Average.**' Then click on '**Bal**' and press '**Max.**' Please refer to the screenshot on next page.



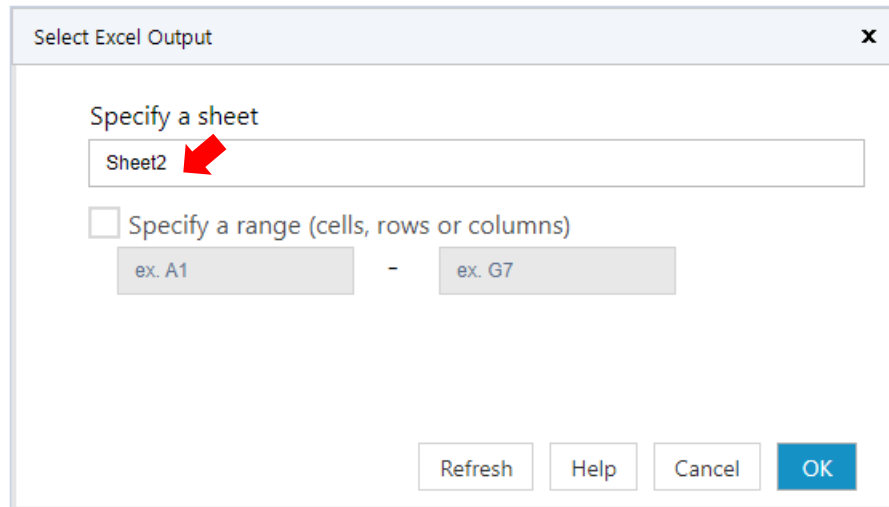
Run the workflow, and you will see the aging analysis.

Results - Summarize (11) - Output

5 of 5 Fields | Cell Viewer | 4 records displayed

Record	Classification	Sum_Bal.	Count	Avg_Bal.	Max_Bal.
1	30-59 days	2457.36	29	84.736552	152.14
2	over 90 days	3338.38	15	222.558667	1566.12
3	current	3189.24	43	74.168372	151.59
4	60-89 days	1637.12	20	81.856	136.38

- E) Now we want to save the data to a new sheet (**Sheet2**) in the same Excel workbook as Sheet 1. To do this, connect the Output tool to the Summarize tool and configure the settings so that the output file is the same as before; the output will look like the one in *Figure 2*.



Select Excel Output

Specify a sheet

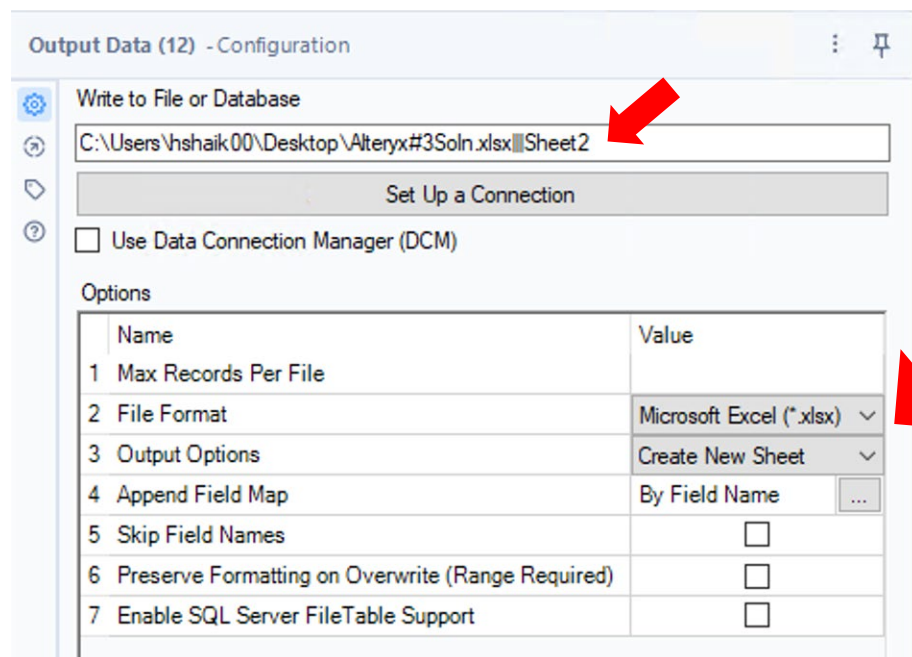
Sheet2

☐ Specify a range (cells, rows or columns)

ex. A1 - ex. G7

Refresh Help Cancel OK

- F) The output options can stay on **'Create New Sheet'** because the old one is being dropped in the previous output tool. This works because there is no old sheet with this same tab name because it was dropped.



Output Data (12) - Configuration

Write to File or Database

C:\Users\hshaik00\Desktop\Alteryx#3Soln.xlsx\Sheet2

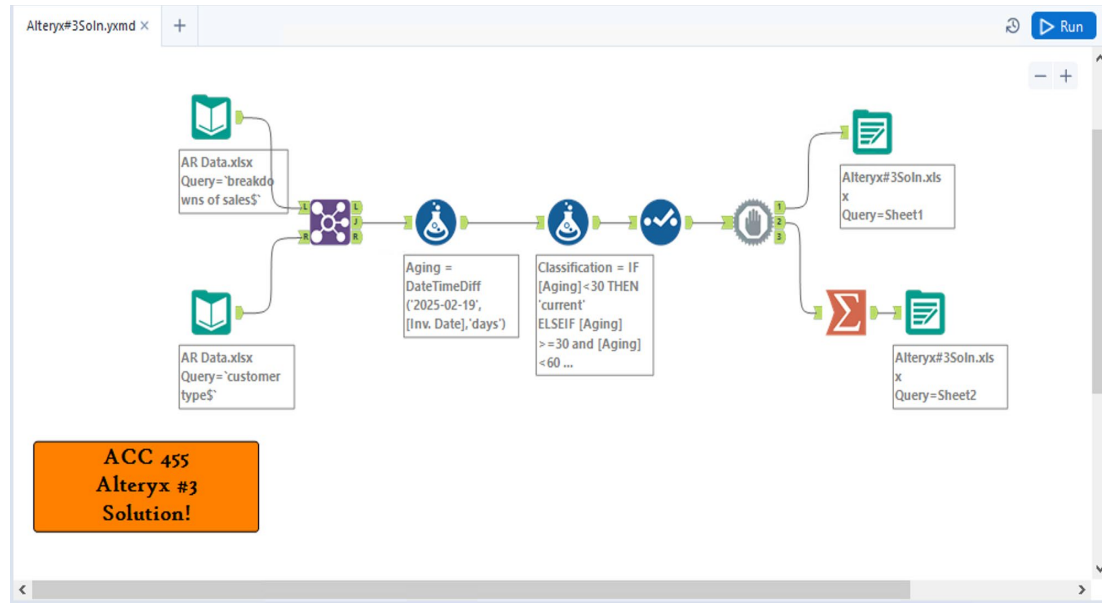
Set Up a Connection

☐ Use Data Connection Manager (DCM)

Options

Name	Value
1 Max Records Per File	
2 File Format	Microsoft Excel (*.xlsx)
3 Output Options	Create New Sheet
4 Append Field Map	By Field Name
5 Skip Field Names	☐
6 Preserve Formatting on Overwrite (Range Required)	☐
7 Enable SQL Server FileTable Support	☐

- G) Run the workflow and look at the aging analysis. **Output—Figure 2.**



Classification					
	A	B	C	D	E
1	Classification	Sum_Bal.	Count	Avg_Bal.	Max_Bal.
2	30-59 days	2457.36	29	84.73655	152.14
3	over 90 days	3338.38	15	222.5587	1566.12
4	current	3189.24	43	74.16837	151.59
5	60-89 days	1637.12	20	81.856	136.38

Figure 2. An Aging Analysis Showing the Sums, Averages, and Counts of Account Balances for the Data in Figure 1

Part 2: Creating a Two-Dimensional Aging Analysis

When performing audits of accounts receivable data, it may also be useful to create a two-dimensional table that displays account balances classified by both transaction age and customer.

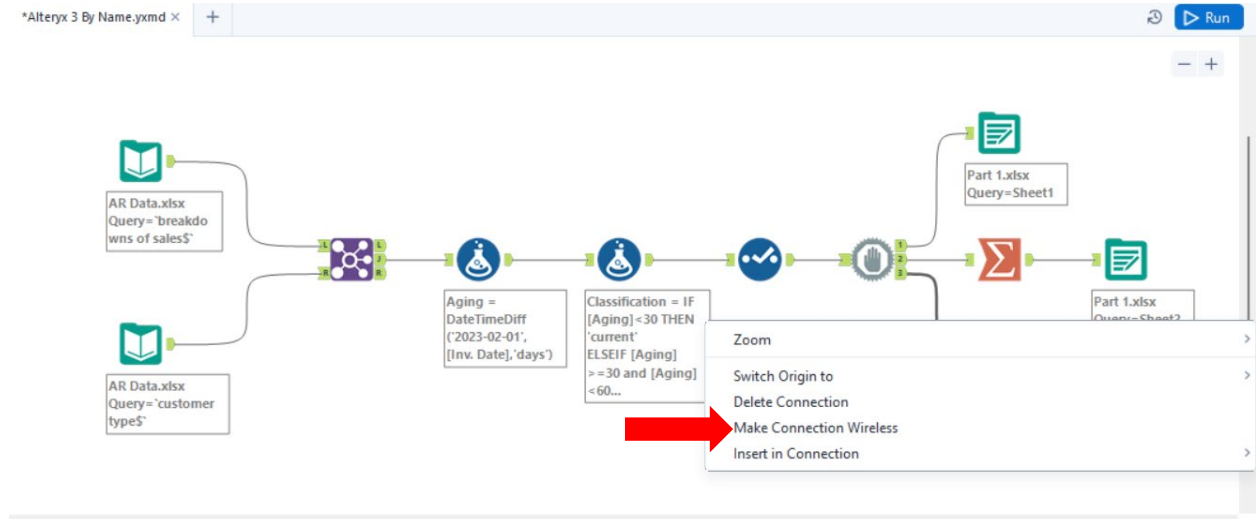
Figure 3 displays the desired analysis.

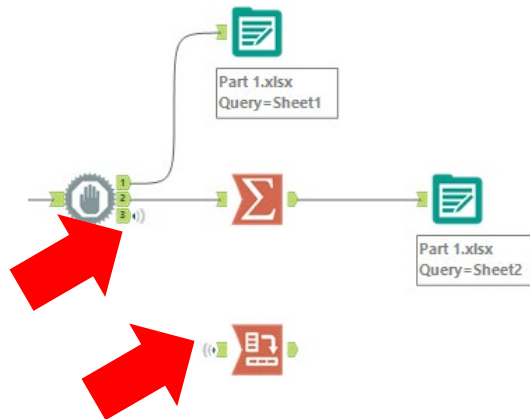
	A	B	C	D	E	F	G
	Customer	Sum_30_59_days	Sum_60_89_days	Sum_current	Sum_over_90_days	Total	
2	2			793.54		793.54	
3	4	552.59	431.55	286.54	98.1	1368.78	
4	6				215.32	215.32	
5	1			295.46		295.46	
6	10				2521.66	2521.66	
7	7			156.45		156.45	
8	9	71.04	354.99	278.46		704.49	
9	8	599.61	174.24	353.5	279.01	1406.36	
10	5	484.25	362.61	618.46		1465.32	
11	3	749.87	313.73	406.83	224.29	1694.72	
12		2457.36	1637.12	3189.24	3338.38	10622.1	
13							

Figure 3. A Two-Dimension Aging Analysis

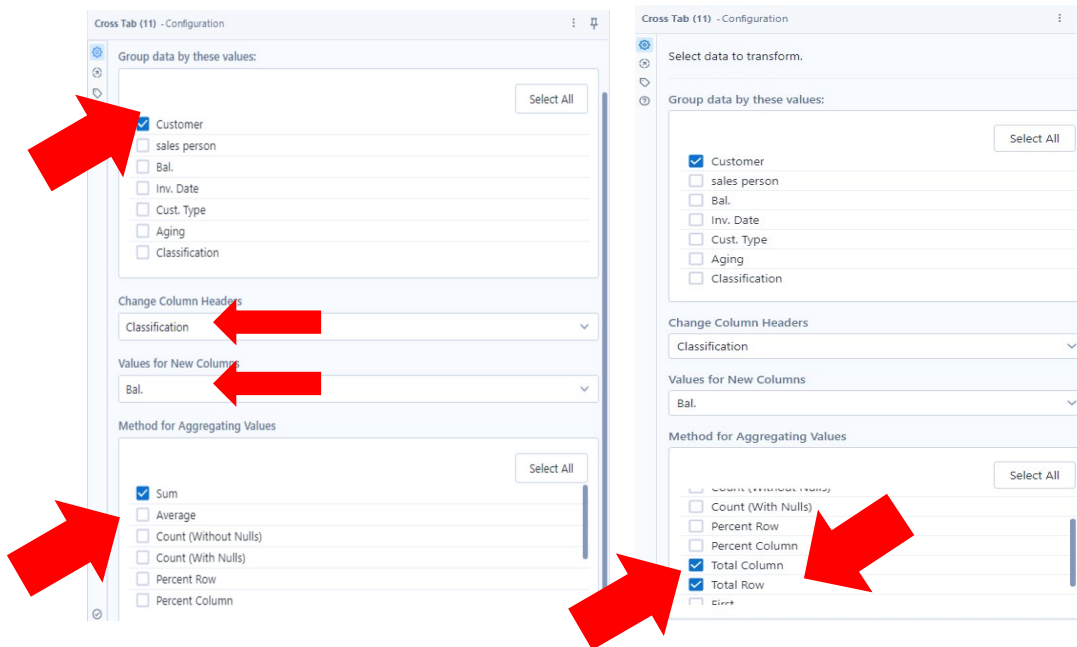
The following are the steps to create the two-dimensional pivot table shown in Figure 3:

- A) Drag the **Cross Tab** tool from the **'Transform'** toolbar or type **'Cross Tab'** in the search field onto the canvas and connect it to **#3** of the Block Until Done tool. Then right-click on the link between them and choose 'make the connection wireless'.





- B) In the configuration tab, group the data by **'Customer,'** change the column headers to **'Classification,'** and values for new columns should be based on **'Bal.'** Under Method for Aggregating Values, click on **'Sum,' 'Total Column,'** and **'Total Row.'**



- C) Run the workflow. Take a look at the new two-dimensional table!

Results - Cross Tab (13) - Output

6 of 6 Fields ☒ ☐ Cell Viewer 11 records displayed ✖ ✔



Record	Customer	Sum_30_59_days	Sum_60_89_days	Sum_current	Sum_over_90_days	Total
1	2	[Null]	[Null]	793.54	[Null]	793.54
2	4	552.59	431.55	286.54	98.1	1368.78
3	6	[Null]	[Null]	[Null]	215.32	215.32
4	1	[Null]	[Null]	295.46	[Null]	295.46
5	10	[Null]	[Null]	[Null]	2521.66	2521.66
6	7	[Null]	[Null]	156.45	[Null]	156.45
7	9	71.04	354.99	278.46	[Null]	704.49
8	8	599.61	174.24	353.5	279.01	1406.36
9	5	484.25	362.61	618.46	[Null]	1465.32
10	3	749.87	313.73	406.83	224.29	1694.72
11	[Null]	2457.36	1637.12	3189.24	3338.38	10622.1

- D) Now we want to write the table on a new Excel sheet in the same workbook as earlier. We will save it as **Sheet 3**.

Select Excel Output ✖

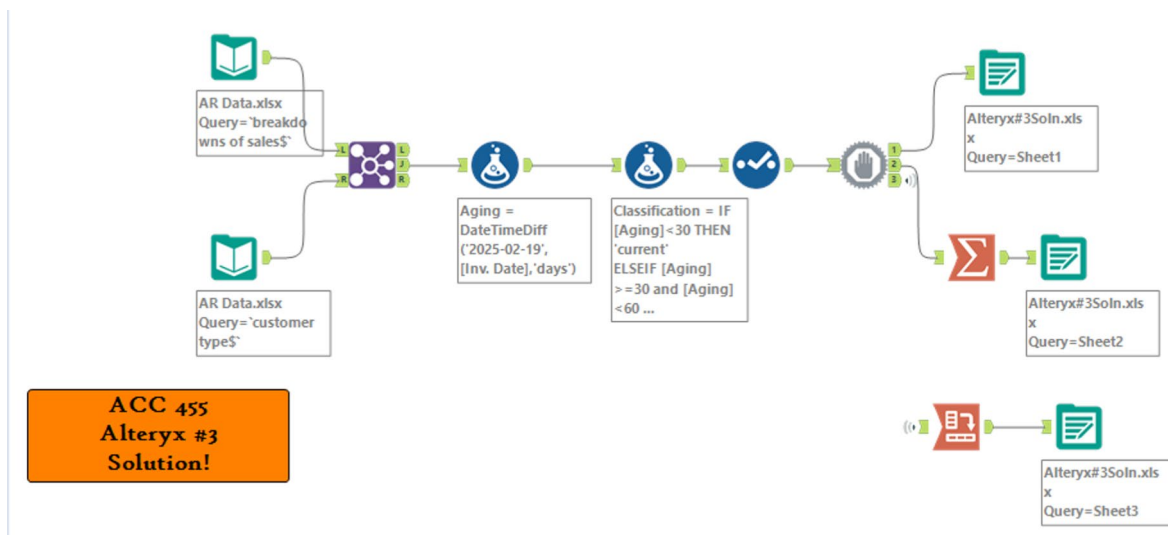
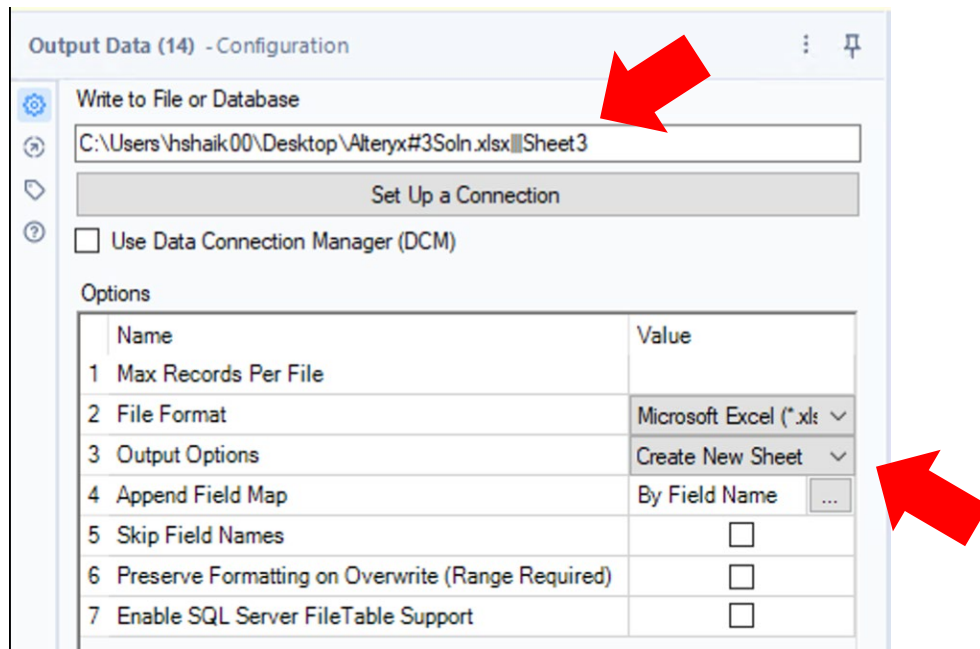
Specify a sheet ➔

Sheet3

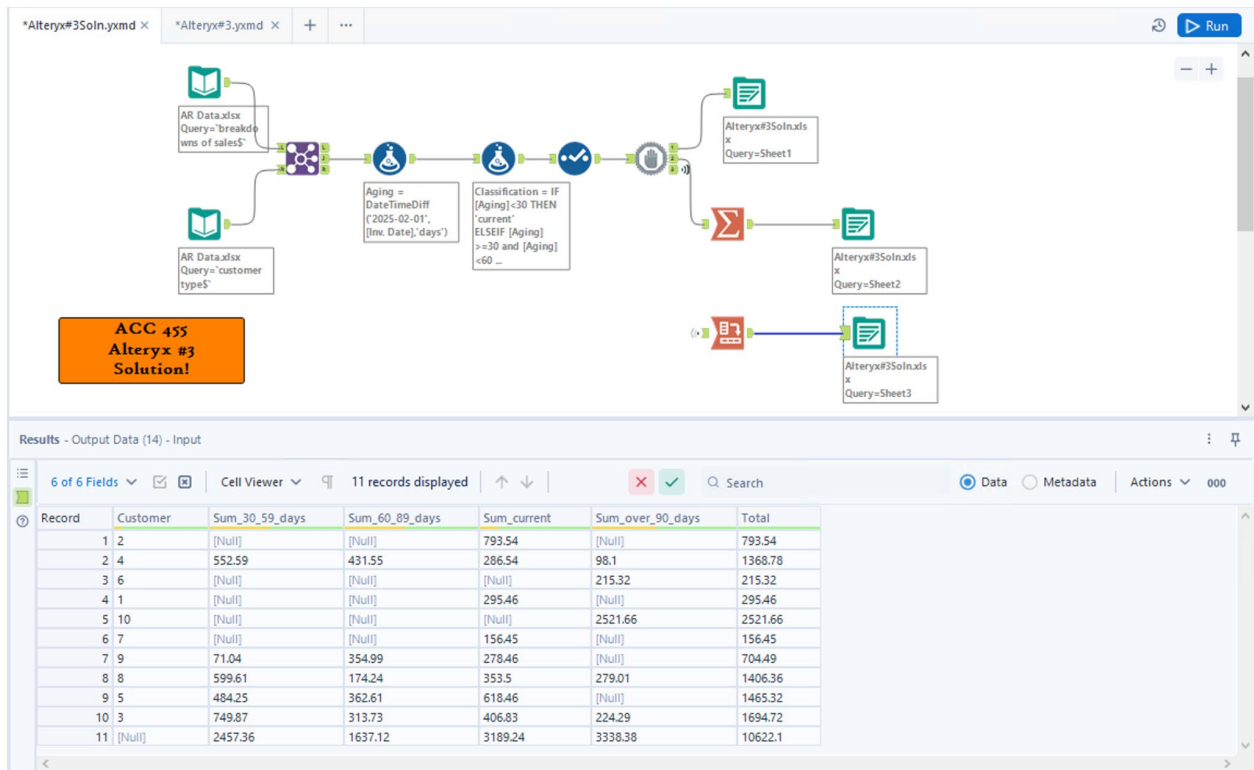
☐ Specify a range (cells, rows or columns)

ex. A1 - ex. G7

Connect the **Output tool** to the Cross Tab tool and configure the settings so that the output file is the **same output file** as before. The output options can stay on **'Create New Sheet.'**



E) Run the workflow. Take a look at the new Excel sheet; it should look the way it does in Figure 3.



With the data for the table properly specified and formatted, it is now easy to make some observations about this set of transactions.

Part 3 (Exercise): Creating a Three-Dimensional Aging Analysis

If an auditor wishes to classify receivables in some way other than by Customer and Aging Category, he/she needs only to create a new pivot table using alternate data fields. To illustrate, perhaps the auditor wishes to tabulate the amounts by both age classification and customer type. In the initial dataset of **Figure 1**, for example, there are three types of customers with codes X, Y, and Z.

Do the aged accounts receivables differ by customer type?

To find out, an auditor can create the pivot table in **Figure 4**. Here, the aging categories form one dimension of the table, and the customer codes form the second dimension. The column variable is now 'Cust.Type' and the row variable is now 'Classification'. Please create a pivot just as in **Figure 4**.

Hint: Drop another Cross Tab tool onto the canvas and connect it to **#3 of the 'Block until Done'** tool. Make the connection wireless. Then configure the settings (you should be able to figure this out on your own), and when you run the workflow, make sure that your table looks like the table in Figure 4. Then use the Output Data tool to save the work onto a new Excel sheet, **Sheet 4**. Make sure that the table in Sheet 4 looks the same as the table in Figure 4 (it's okay if your rows are in a different order; just make sure that your numbers are correct).

Results - Output Data (16) - Input

5 of 5 Fields ☒ ☐ | Cell Viewer ☐ | 5 records displayed | ↑ ↓

Record	Classification	Sum_X	Sum_Y	Sum_Z	Total
1	30-59 days	[Null]	1786.71	670.65	2457.36
2	over 90 days	[Null]	537.71	2800.67	3338.38
3	current	1089	1468.28	631.96	3189.24
4	60-89 days	[Null]	1107.89	529.23	1637.12
5	[Null]	1089	4900.59	4632.51	10622.1

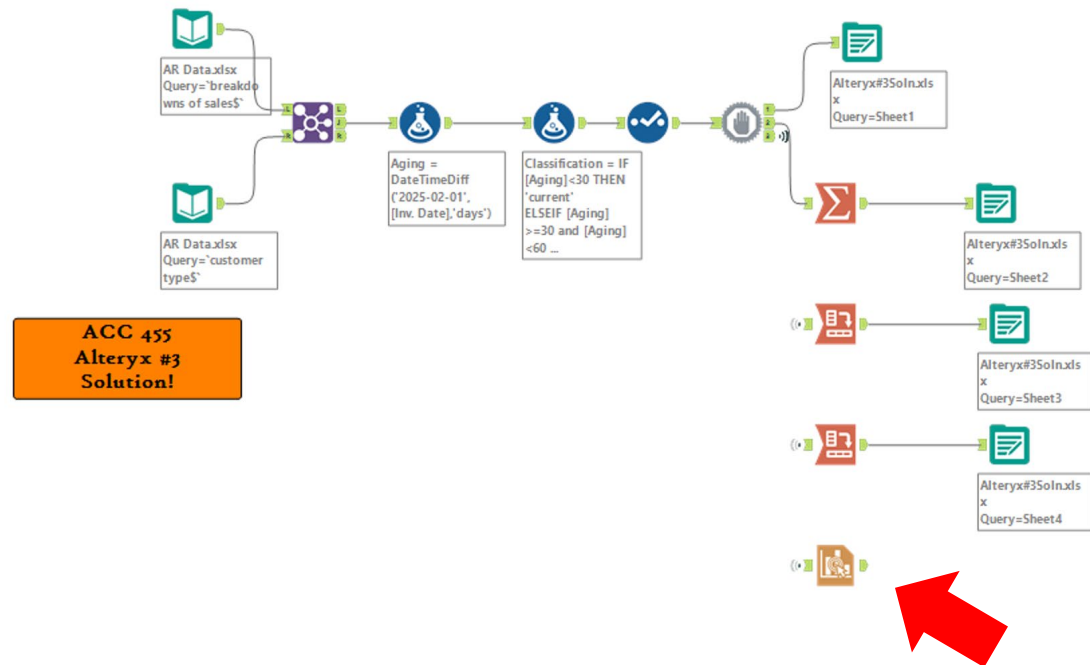
	A	B	C	D	E
1	Classification	Sum_X	Sum_Y	Sum_Z	Total
2	30-59 days		1786.71	670.65	2457.36
3	over 90 days		537.71	2800.67	3338.38
4	current	1089	1468.28	631.96	3189.24
5	60-89 days		1107.89	529.23	1637.12
6		1089	4900.59	4632.51	10622.1

Figure 4. A Three-Dimensional Aging Analysis

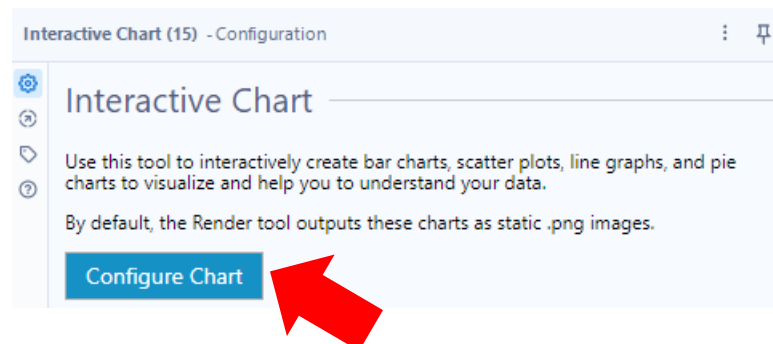
Part 4: Creating a Chart

Now we can create a bar chart to visualize the data.

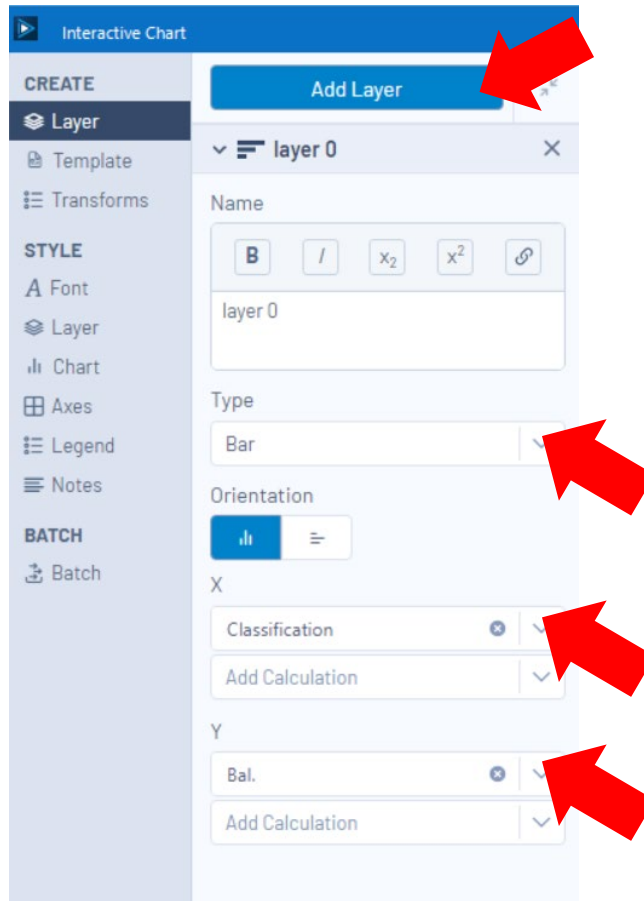
- A) We will use the **Interactive Chart tool** to do this. Drag the **Interactive Chart tool** from the **Reporting Tool Tab** onto the canvas and connect it to **#3** of the **Block Until Done** tool. Then make the connection wireless.



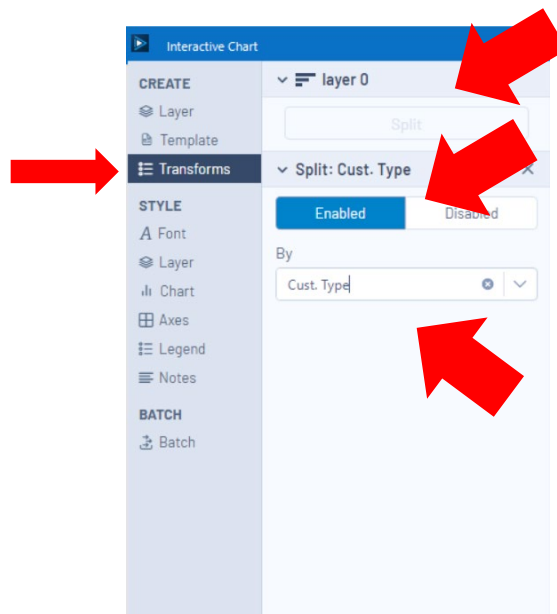
- B) Click on 'Configure Chart,' and a new page should open up. Click on 'Add layer' and change the type to Bar.

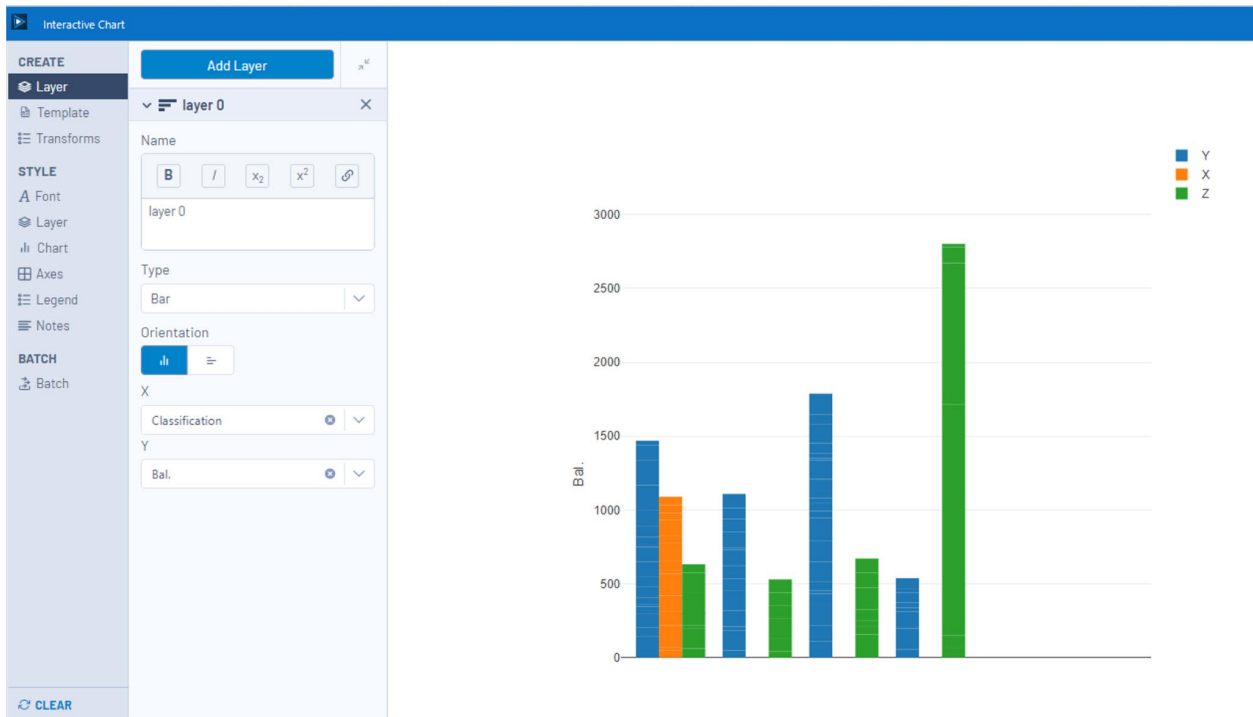


- C) For X, select the column to be based on 'Classification', and for Y, the column should be based on 'Bal.'



D) Then click on Transforms in the top left and use the **Split** button to split by 'Cust. Type.' Notice how the bar chart changes after this.





- E) To output this data, we will use the Render tool. Drag the Render tool onto the canvas and connect it to the Interactive Chart tool.



- F) In the **configuration tab**, change the output mode to 'Temporary Excel 2007 Spreadsheet.' Then run the workflow. When you click the link in the message tab in the result panel and open Excel, the chart should look the way it does in **Figure 5**.



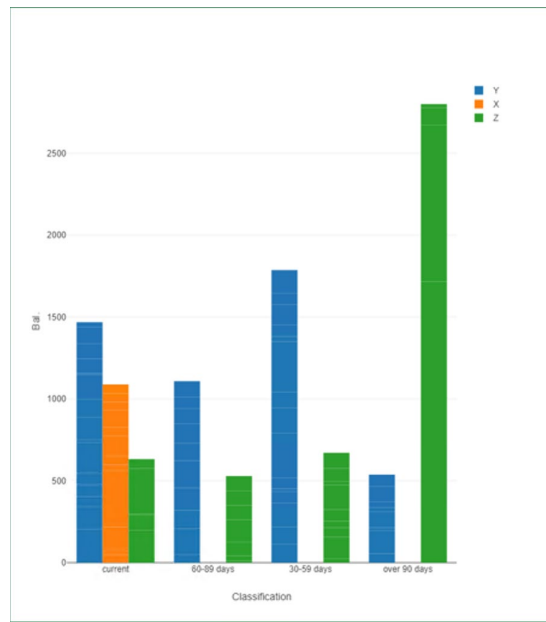
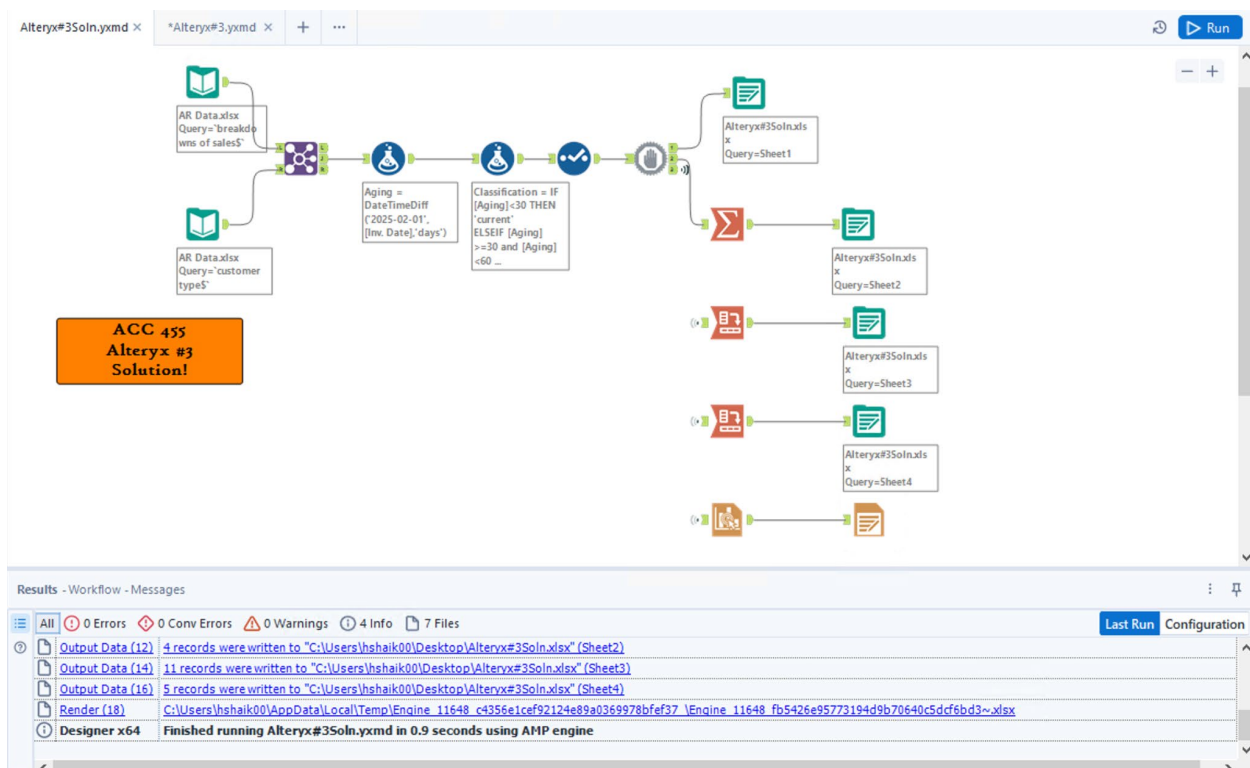


Figure 5. A Three-Dimension Aging Analysis Using a Bar Chart

Final Output:



The End!